Organisms and their environment

Mark scheme and worked solutions for 30 questions: 15 multiple choice, 15 structured.

1 ? Part 1

Core · Energy flow · [1]

Answer: B

Explanation

Sunlight supplies the principal energy input, captured by producers. Minerals and carbon dioxide supply materials rather than the main energy input.

2 ? Part 1

Core · Food chains and food webs · [1]

Answer: D

Explanation

A producer makes organic nutrients, usually through photosynthesis. Rabbits and hawks consume organisms; the fungus is a decomposer.

3 ? Part 1

Core · Food chains and food webs · [1]

Answer: A

Explanation

The grasshopper is a primary consumer feeding on the producer. The lizard eats that primary consumer and is therefore secondary.

4 ? Part 1

Core · Food chains and food webs · [1]

Answer: C

Explanation

Decomposers obtain energy from dead or waste organic material. Photosynthesis identifies producers and grazing identifies herbivores.

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5 ? Part 1

Core | Food chains and food webs | [1]

Answer: D

Explanation

Carnivores obtain energy by eating animals. Herbivores eat plants; producers make their own organic nutrients.

6 ? Part 1

Core | Food chains and food webs | [1]

Answer: B

Explanation

A food web links several chains and can show multiple food sources. It includes producers and does not imply energy recycling.

7 ? Part 1

Core | Food chains and food webs | [1]

Answer: C

Explanation

Removing predators reduces mortality of their small-fish prey, allowing those fish to increase. The other outcomes do not follow from the given relationships.

8 ? Part 1

Core | Food chains and food webs | [1]

Answer: A

Explanation

Biomass takes the amount of living material into account. Counts treat a tree and a small herbivore as one each; biomass is not itself an energy-flow rate.

9 ? Part 1

Supplement | Food chains and food webs | [1]

Answer: B

Explanation

Consumers respire and energy is eventually transferred as heat. Animal muscle cells do not photosynthesise, and energy is not all returned to producers.

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10 ? Part 1 Supplement Food chains and food webs [1]

Answer: C

Explanation

Energy transfer is inefficient, leaving less available at higher levels. This limits the populations that a higher trophic level can support.

11 ? Part 1 Supplement Food chains and food webs [1]

Answer: D

Explanation

Cattle use energy in respiration and do not convert all crop material into edible tissue. Direct consumption avoids an additional transfer step.

12 ? Part 1 Supplement Nutrient cycles [1]

Answer: A

Explanation

Nitrogen fixation converts atmospheric nitrogen into compounds. Denitrification releases nitrogen gas, while evaporation and carbohydrate combustion do not fix it.

13 ? Part 1 Core Populations [1]

Answer: C

Explanation

A population contains organisms of one species in the same area at the same time. A community includes different species.

14 ? Part 1 Core Populations [1]

Answer: A

Explanation

A community includes all the populations of different species. One species is a population; physical conditions alone are not a community.

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15 ? Part 1 Core Populations [1]

Answer: B

Explanation

Disease can increase mortality or reduce reproduction and therefore limit population growth. It does not create resources or make animals producers.

16 ? Part 1 Core Energy flow [4]

Mark scheme

(a) **[2]**

- Green plants absorb light energy from the Sun. (1)
- Photosynthesis stores energy as chemical energy in organic nutrients. (1)

(b) **[2]**

- Feeding transfers chemical energy between organisms. (1)
- Energy is eventually transferred to the environment, for example as heat from respiration. (1)

Worked solution

(a) Green plants absorb light energy from the Sun. Photosynthesis stores energy as chemical energy in organic nutrients.

(b) Feeding transfers chemical energy between organisms. Energy is eventually transferred to the environment, for example as heat from respiration.

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17 ? Part 1 Core Food chains and food webs [4]

Mark scheme

(a) **[2]**

- Pondweed → snail. (1)
- Snail → fish → heron, completing the chain in this order. (1)

(b) **[2]**

- Pondweed is the producer. (1)
- The heron is the tertiary consumer. (1)

Worked solution

(a) Pondweed → snail. Snail → fish → heron, completing the chain in this order.

(b) Pondweed is the producer. The heron is the tertiary consumer.

18 ? Part 1

Core | Food chains and food webs | [4]

Mark scheme

(a) **[2]**

- Owls have less available food. (1)
- Owl survival or reproduction may decrease, reducing their population. (1)

(b) **[2]**

- Foxes can also eat rabbits. (1)
- This alternative food source reduces their dependence on voles. (1)

Worked solution

(a) Owls have less available food. Owl survival or reproduction may decrease, reducing their population.

(b) Foxes can also eat rabbits. This alternative food source reduces their dependence on voles.

19 ? Part 1

Core | Food chains and food webs | [4]

Mark scheme

(a) **[2]**

- It obtains energy by feeding on other organisms. (1)
- Those organisms are plants, so it is a herbivore. (1)

(b) **[2]**

- It obtains energy by eating animals. (1)
- It is a secondary consumer in the leaf → hare → wildcat chain. (1)

Worked solution

(a) It obtains energy by feeding on other organisms. Those organisms are plants, so it is a herbivore.

(b) It obtains energy by eating animals. It is a secondary consumer in the leaf → hare → wildcat chain.

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20 ? Part 1

Core | Food chains and food webs | [4]

Mark scheme

(a) **[2]**

- They obtain energy from dead organic material. (1)
- They are decomposers. (1)

(b) **[2]**

- Plants make organic nutrients through photosynthesis. (1)
- They use light energy rather than obtaining all energy by consuming dead material. (1)

Worked solution

(a) They obtain energy from dead organic material. They are decomposers.

(b) Plants make organic nutrients through photosynthesis. They use light energy rather than obtaining all energy by consuming dead material.

21 ? Part 1

Core | Food chains and food webs | [4]

Mark scheme

(a) **[2]**

- Counting gives each tree and each insect equal weight. (1)
- It ignores the much greater amount of living material in one tree. (1)

(b) **[2]**

- Biomass accounts for differences in organism size. (1)
- It better represents the amount of material available at each trophic level. (1)

Worked solution

(a) Counting gives each tree and each insect equal weight. It ignores the much greater amount of living material in one tree.

(b) Biomass accounts for differences in organism size. It better represents the amount of material available at each trophic level.

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22 ? Part 1

Supplement | Food chains and food webs | [4]

Mark scheme

(a) **[2]**

- It represents the energy available to support organisms at each level. (1)
- It measures transfer over time rather than just a snapshot of stored biomass. (1)

(b) **[2]**

- Some energy is used in respiration and transferred as heat. (1)
- Some material is not eaten or is not digested and absorbed. (1)

Worked solution

(a) It represents the energy available to support organisms at each level. It measures transfer over time rather than just a snapshot of stored biomass.

(b) Some energy is used in respiration and transferred as heat. Some material is not eaten or is not digested and absorbed.

23 ? Part 1

Supplement | Food chains and food webs | [4]

Mark scheme

(a) **[2]**

- 2700 divided by 18 000 gives 0.15. (1)
- Multiplying by 100 gives 15%. (1)

(b) **[2]**

- Not all plant material is eaten or digested. (1)
- Herbivores use absorbed energy in respiration, transferring energy to the environment. (1)

Worked solution

(a) 2700 divided by 18 000 gives 0.15. Multiplying by 100 gives 15%.

(b) Not all plant material is eaten or digested. Herbivores use absorbed energy in respiration, transferring energy to the environment.

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24 ? Part 1 Supplement Food chains and food webs [4]

Mark scheme

(a) **[2]**

- It has fewer trophic transfers. (1)
- It avoids energy used by livestock in respiration and lost in uneaten or waste material. (1)

(b) **[2]**

- Each transfer passes on only a fraction of the available energy. (1)
- Eventually too little energy remains to support a further substantial consumer population. (1)

Worked solution

(a) It has fewer trophic transfers. It avoids energy used by livestock in respiration and lost in uneaten or waste material.

(b) Each transfer passes on only a fraction of the available energy. Eventually too little energy remains to support a further substantial consumer population.

25 ? Part 1 Core Food chains and food webs [4]

Mark scheme

(a) **[2]**

- The introduced insect reduces the birds' food supply. (1)
- Increased competition may reduce bird survival and reproduction. (1)

(b) **[2]**

- Fewer birds provide less food for their predator. (1)
- The predator population may then decrease or switch to other prey. (1)

Worked solution

(a) The introduced insect reduces the birds' food supply. Increased competition may reduce bird survival and reproduction.

(b) Fewer birds provide less food for their predator. The predator population may then decrease or switch to other prey.

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26 ? Part 1 Core Nutrient cycles [4]

Mark scheme

(a) **[2]**

- Plants absorb carbon dioxide during photosynthesis. (1)
- Herbivores obtain carbon compounds by feeding on plants. (1)

(b) **[2]**

- Respiration by plants, animals or decomposers. (1)
- Combustion of wood or fossil fuels. (1)

Worked solution

(a) Plants absorb carbon dioxide during photosynthesis. Herbivores obtain carbon compounds by feeding on plants.

(b) Respiration by plants, animals or decomposers. Combustion of wood or fossil fuels.

27 ? Part 1 Supplement Nutrient cycles [4]

Mark scheme

(a) **[2]**

- Decomposers convert nitrogen in proteins to ammonium ions. (1)
- Nitrifying bacteria convert ammonium compounds to nitrate ions that plants can absorb. (1)

(b) **[2]**

- Nitrogen fixation converts nitrogen gas to nitrogen compounds. (1)
- Denitrification converts nitrate ions to nitrogen gas, returning it to the atmosphere. (1)

Worked solution

(a) Decomposers convert nitrogen in proteins to ammonium ions. Nitrifying bacteria convert ammonium compounds to nitrate ions that plants can absorb.

(b) Nitrogen fixation converts nitrogen gas to nitrogen compounds. Denitrification converts nitrate ions to nitrogen gas, returning it to the atmosphere.

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28 ? Part 1 Core Populations [4]

Mark scheme

(a) **[2]**

- The reed population is the organisms of that one species in the area at the same time. (1)
- The community contains all populations of different species in the wetland. (1)

(b) **[2]**

- It includes a community and its physical environment. (1)
- The organisms and the environment interact with one another. (1)

Worked solution

(a) The reed population is the organisms of that one species in the area at the same time. The community contains all populations of different species in the wetland.

(b) It includes a community and its physical environment. The organisms and the environment interact with one another.

29 ? Part 1 Core Populations [4]

Mark scheme

(a) **[2]**

- Rabbits compete more for limited food. (1)
- Less food can reduce survival or reproduction. (1)

(b) **[2]**

- More foxes can kill more rabbits. (1)
- This increases mortality and reduces net population growth. (1)

Worked solution

(a) Rabbits compete more for limited food. Less food can reduce survival or reproduction.

(b) More foxes can kill more rabbits. This increases mortality and reduces net population growth.

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30 ? Part 1 Supplement Populations [4]

Mark scheme

(a) **[2]**

- Initially the bacteria adapt to the conditions and there are few reproducing cells. (1)
- Later many bacteria reproduce while nutrients and other resources are plentiful. (1)

(b) **[2]**

- Nutrients become limiting and wastes accumulate. (1)
- Birth and death rates become approximately equal in the stationary phase. (1)

Worked solution

(a) Initially the bacteria adapt to the conditions and there are few reproducing cells. Later many bacteria reproduce while nutrients and other resources are plentiful.

(b) Nutrients become limiting and wastes accumulate. Birth and death rates become approximately equal in the stationary phase.

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